3 Hardware

3.1 Computer architecture

Candidates should be able to:	Notes and guidance
1 (a) Understand the role of the central processing unit (CPU) in a computer	• The CPU processes instructions and data that are input into the computer so that the result can be output
(b) Understand what is meant by a microprocessor	• A microprocessor is a type of integrated circuit on a single chip
2 (a) Understand the purpose of the components in a CPU, in a computer that has a Von Neumann architecture	• Including: – units: arithmetic logic unit (ALU) and control unit (CU) – registers: program counter (PC), memory address register (MAR), memory data register (MDR), current instruction register (CIR) and accumulator (ACC) – buses: address bus, data bus and control bus
(b) Describe the process of the fetch–decode–execute (FDE) cycle, including the role of each component in the process	• How instructions and data are fetched from random access memory (RAM) into the CPU, how they are processed using each component and how they are then executed • Storing data and addresses into specific registers • Using buses to transmit data, addresses and signals • Using units to fetch, decode and execute data and instructions
3 Understand what is meant by a core, cache and clock in a CPU and explain how they can affect the performance of a CPU	• The number of cores, size of the cache and speed of the clock can affect the performance of a CPU
4 Understand the purpose and use of an instruction set for a CPU	• An instruction set is a list of all the commands that can be processed by a CPU, and the commands are machine code
5 Describe the purpose and characteristics of an embedded system and identify devices in which they are commonly used	• An embedded system is used to perform a dedicated functions. For example in, domestic appliances, cars, security systems, lighting systems or vending machines. This is different to a general purpose computer that is used to perform many different functions. For example in, a personal computer (PC) or a laptop

3.2 Input and output devices

Candidates should be able to:	Notes and guidance
1 Understand what is meant by an input device and why it is required	• Including: – barcode scanner – digital camera – keyboard – microphone – optical mouse – QR code scanner – touch screen (resistive, capacitive and infra-red) – two-dimensional (2D) and three-dimensional (3D) scanners
2 Understand what is meant by an output device and why it is required	• Including: – actuator – digital light processing (DLP) projector – inkjet printer – laser printer – light emitting diode (LED) screen – liquid crystal display (LCD) projector – liquid crystal display (LCD) screen – speaker – 3D printer
3 (a) Understand what is meant by a sensor and the purposes of sensors	• Limited to: – acoustic – accelerometer – flow – gas – humidity – infra-red – level – light – magnetic field – moisture – pH – pressure – proximity – temperature
(b) Identify the type of data captured by each sensor and understand when each sensor would be used, including selecting the most suitable sensor for a given context	

3.3 Data storage

Candidates should be able to:	Notes and guidance
1 Understand what is meant by primary storage	• Primary storage is directly accessed by the CPU • Including the role of: – random access memory (RAM) – read only memory (ROM) • Including why a computer needs both RAM and ROM, and the difference between them
2 Understand what is meant by secondary storage	• Secondary storage is not directly accessed by the CPU and is necessary for more permanent storage of data
3 Describe the operation of magnetic, optical and solid-state (flash memory) storage and give examples of each	• Magnetic storage uses platters which are divided into tracks and sectors. Data is read and written using electromagnets. Including hard disk drive (HDD) • Optical storage uses lasers to create and read pits and lands. Including: CD, DVD and Blu-ray • Solid-state (flash memory) uses NAND or NOR technology. Transistors are used as control gates and floating gates. Including: solid-state drive (SSD), SD card and USB drive
4 Describe what is meant by virtual memory, how it is created and used and why it is necessary	• Pages of data are transferred between RAM and virtual memory when needed
5 Understand what is meant by cloud storage	• Cloud storage can be accessed remotely in comparison to storing data locally
6 Explain the advantages and disadvantages of storing data on the cloud in comparison to storing it locally	• Physical servers and storage are needed to store data in cloud storage

3.4 Network hardware

Candidates should be able to:	Notes and guidance
1 Understand that a computer needs a network interface card (NIC) to access a network	
2 Understand what is meant by, and the purpose of, a media access control (MAC) address, including its structure	• A network interface card is given a MAC address at the point of manufacture • MAC addresses are usually written as hexadecimal • MAC addresses are created using the manufacturer code and the serial code
3 (a) Understand what is meant by, and the purpose of, an internet protocol (IP) address (b) Understand that there are different types of IP address	• An IP address is allocated by the network and it can be static or dynamic • Including the characteristics of, and differences between, IPv4 and IPv6
4 Describe the role of a router in a network	• A router sends data to a specific destination on a network • A router can assign IP addresses • A router can connect a local network to the internet

4 Software

4.1 Types of software and interrupts

Candidates should be able to:	Notes and guidance
1 Describe the difference between system software and application software and provide examples of each	• System software provides the services that the computer requires, including operating system and utility software • Application software provides the services that the user requires
2 Describe the role and basic functions of an operating system	• Including: – managing files – handling interrupts – providing an interface – managing peripherals and drivers – managing memory – managing multitasking – providing a platform for running applications – providing system security – managing user accounts